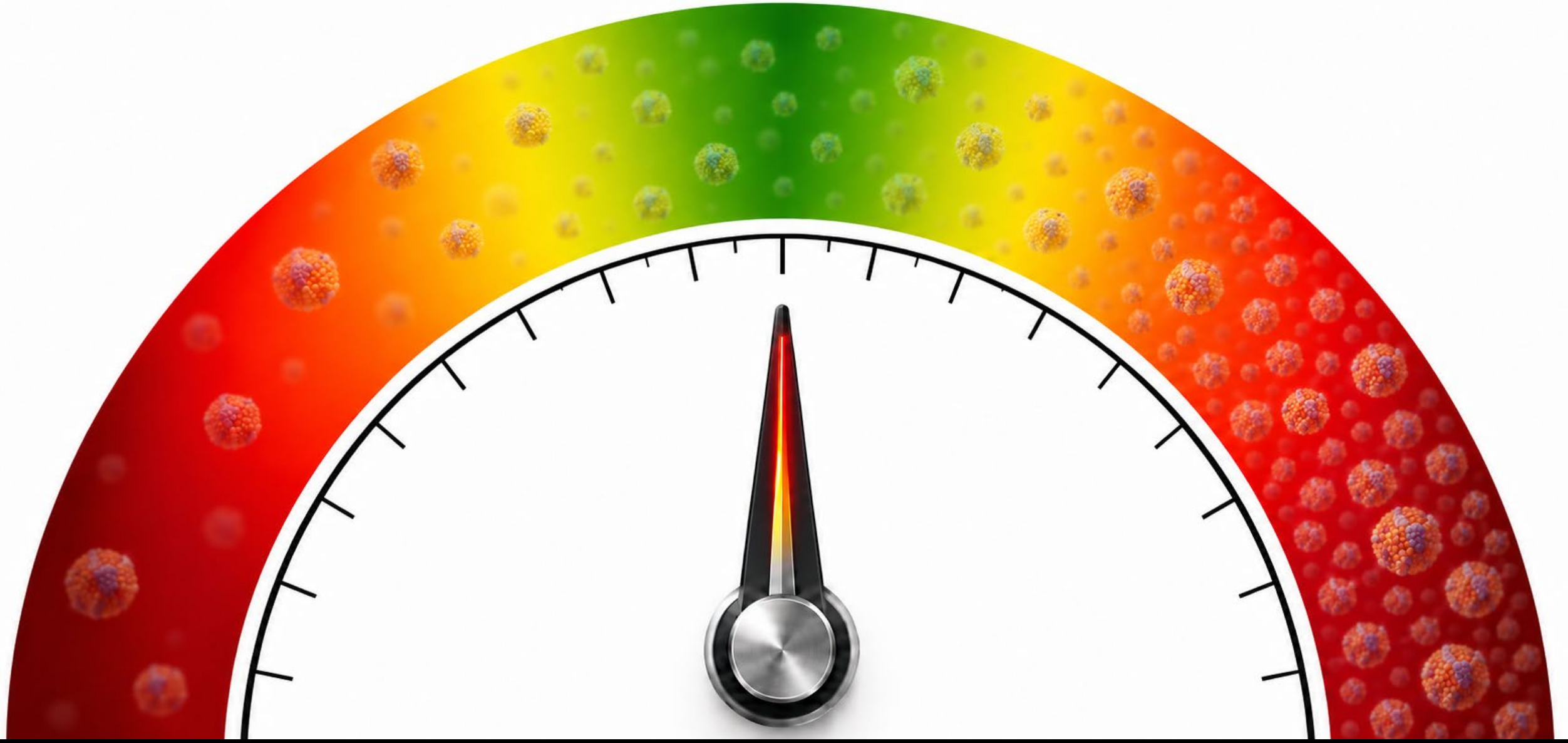
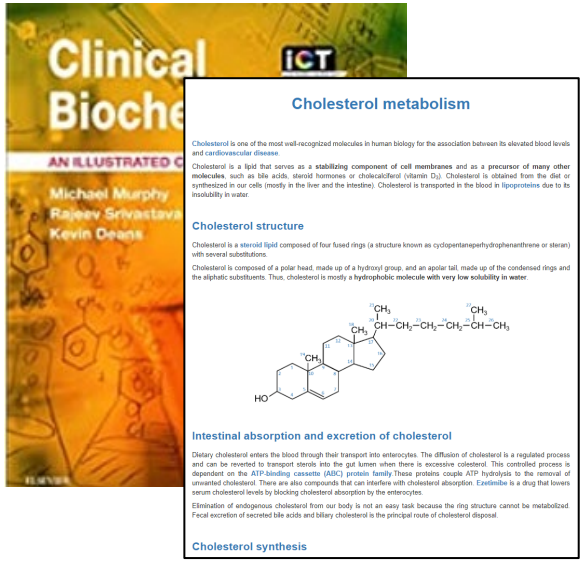


Cholesterol metabolism

Lipid transport

Supporting material



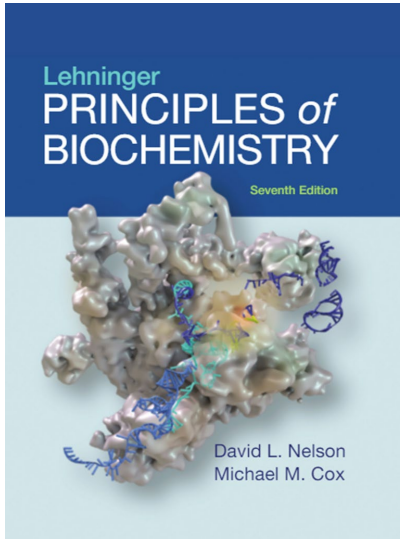
Additional study materials

ADI document:

Cholesterol metabolism

Murphy's CB Chapter 67:

Lipoprotein metabolism



Reading

Lehninger's PoB Chapter 21:

21.4 Cholesterol, steroids, and isoprenoids

Learning goals

- The four phases of cholesterol synthesis
- Regulation of cholesterol synthesis
- Fates of cholesterol molecules
- Lipoprotein metabolism

Cholesterol functions and sources

- Functions 
  - Membranes
  - Precursor of bile acids
  - Precursor of steroid hormones
  - Precursor of vitamin D
- Sources 
  - Diet
  - *De novo* synthesis

Cholesterol synthesis: acetyl-CoA as the carbon source

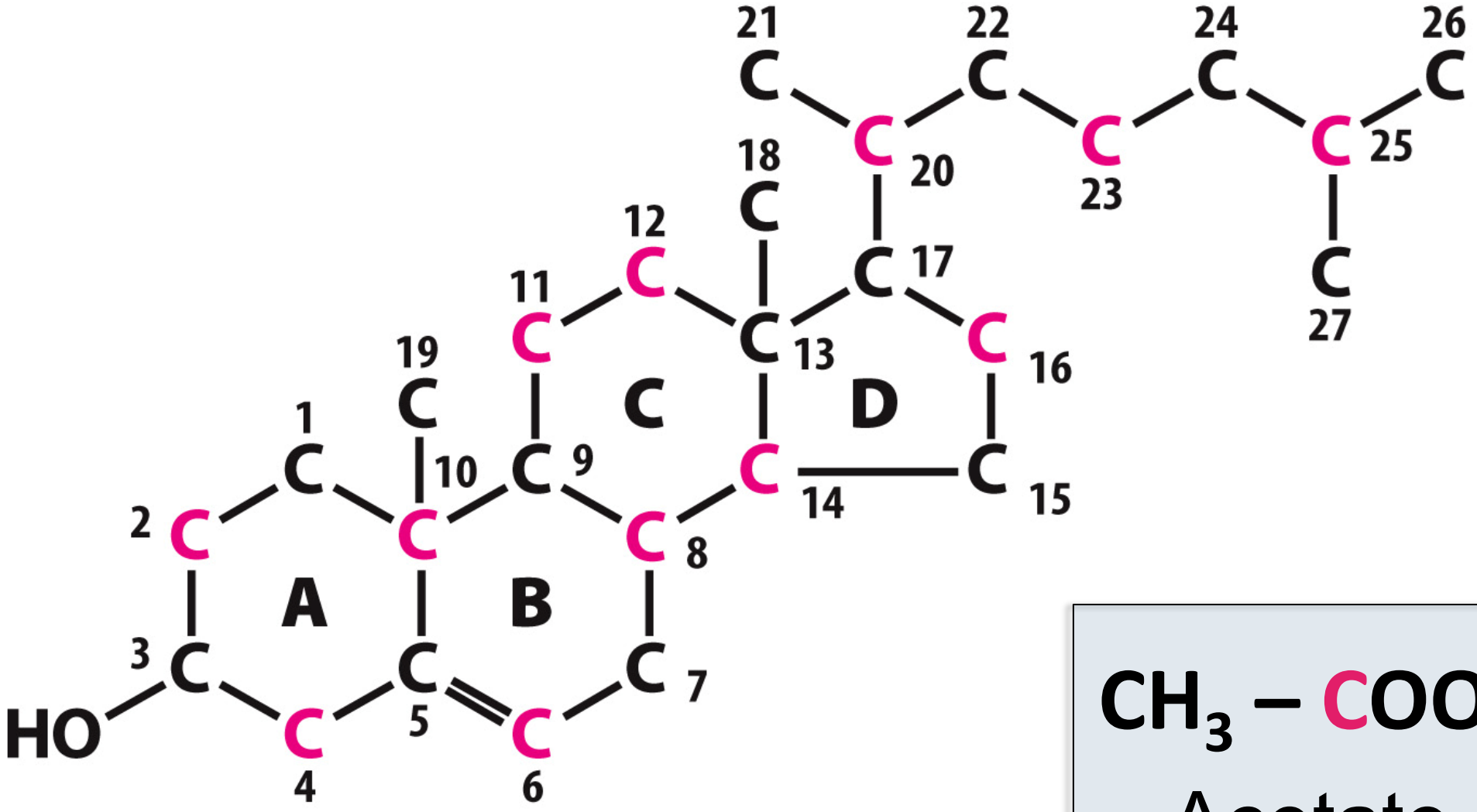


Figure 21-32
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$CH_3 - COO^-$
Acetate

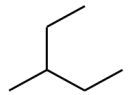
Cholesterol synthesis: phases

1. Acetyl-CoA → Mevalonate

2C x 3

↓

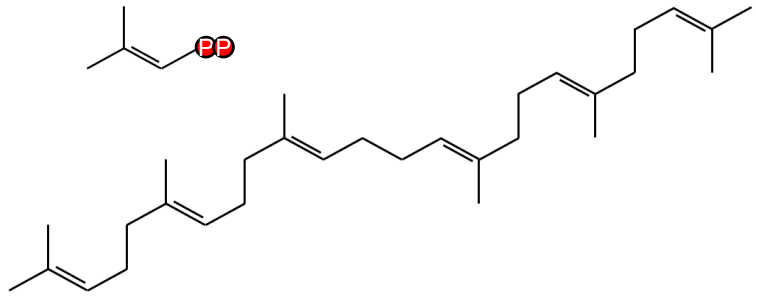
6C



2. Mevalonate → Activated isoprene + CO_2

↓

5C x 6



3. Isoprene units → Squalene

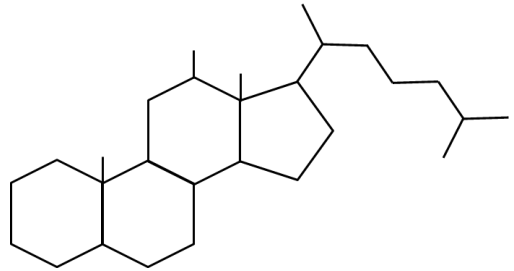
↓

30C

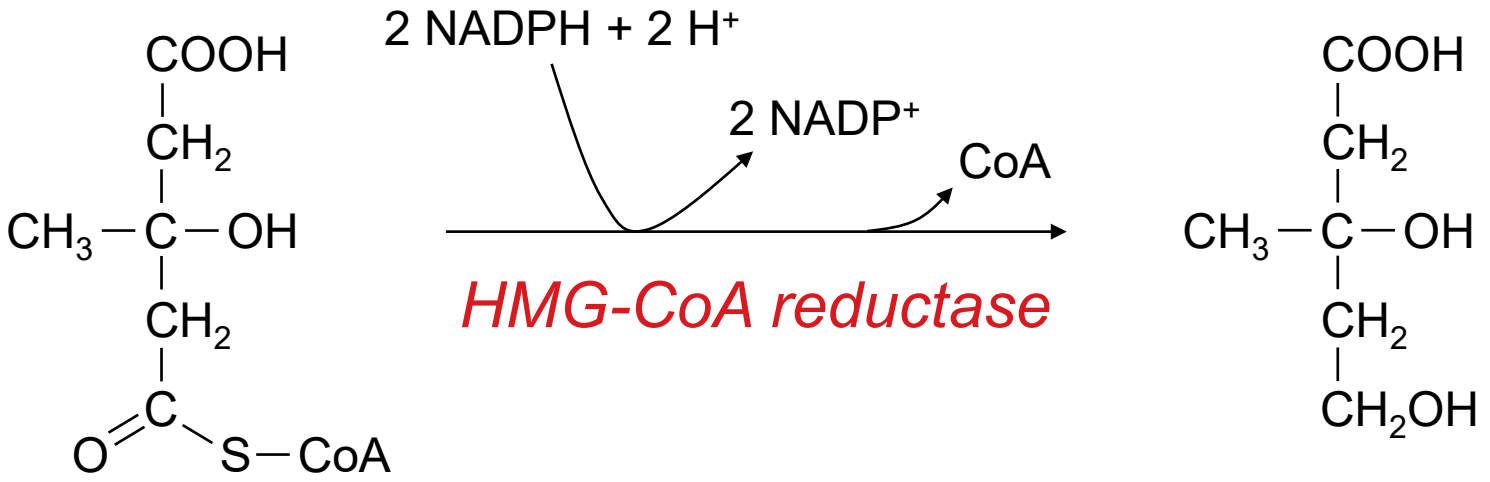
4. Squalene → Cholesterol

↓

27C



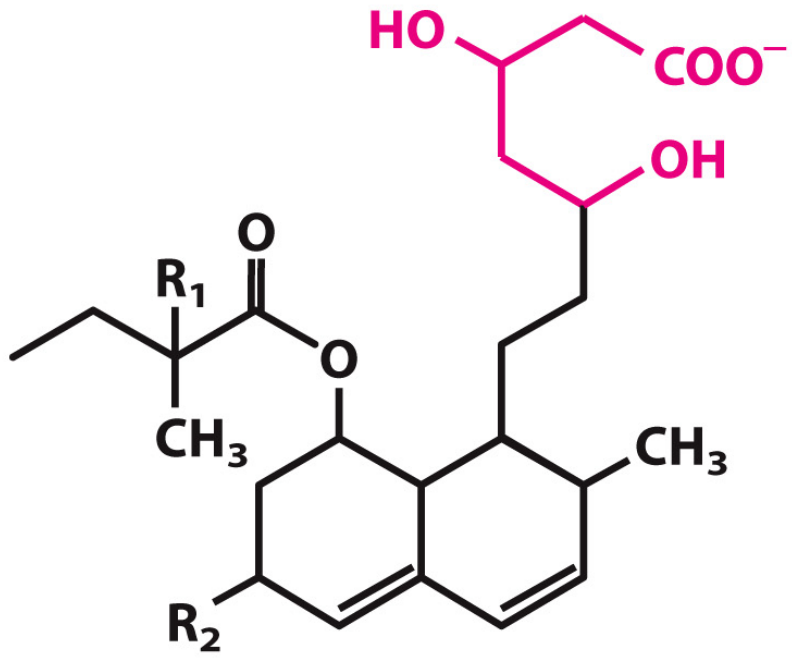
Cholesterol synthesis: formation of mevalonate

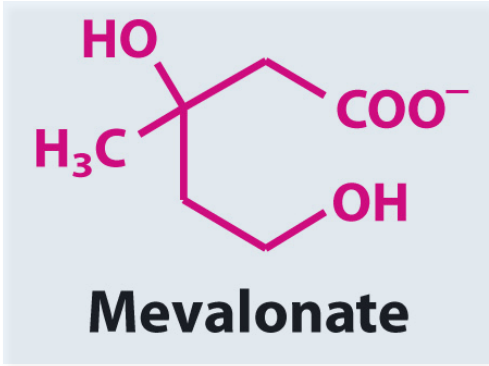


3-hydroxy-3-methylglutaryl coenzyme A (HMG-CoA)

Mevalonate

Statins: competitive inhibitors of *HMG-CoA reductase*





Mevalonate

$R_1 = H$	$R_2 = H$	Compactin
$R_1 = CH_3$	$R_2 = CH_3$	Simvastatin (Zocor)
$R_1 = H$	$R_2 = OH$	Pravastatin (Pravachol)
$R_1 = H$	$R_2 = CH_3$	Lovastatin (Mevacor)

Box 21-3 Figure 1
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Statins: competitive inhibitors of *HMG-CoA reductase*

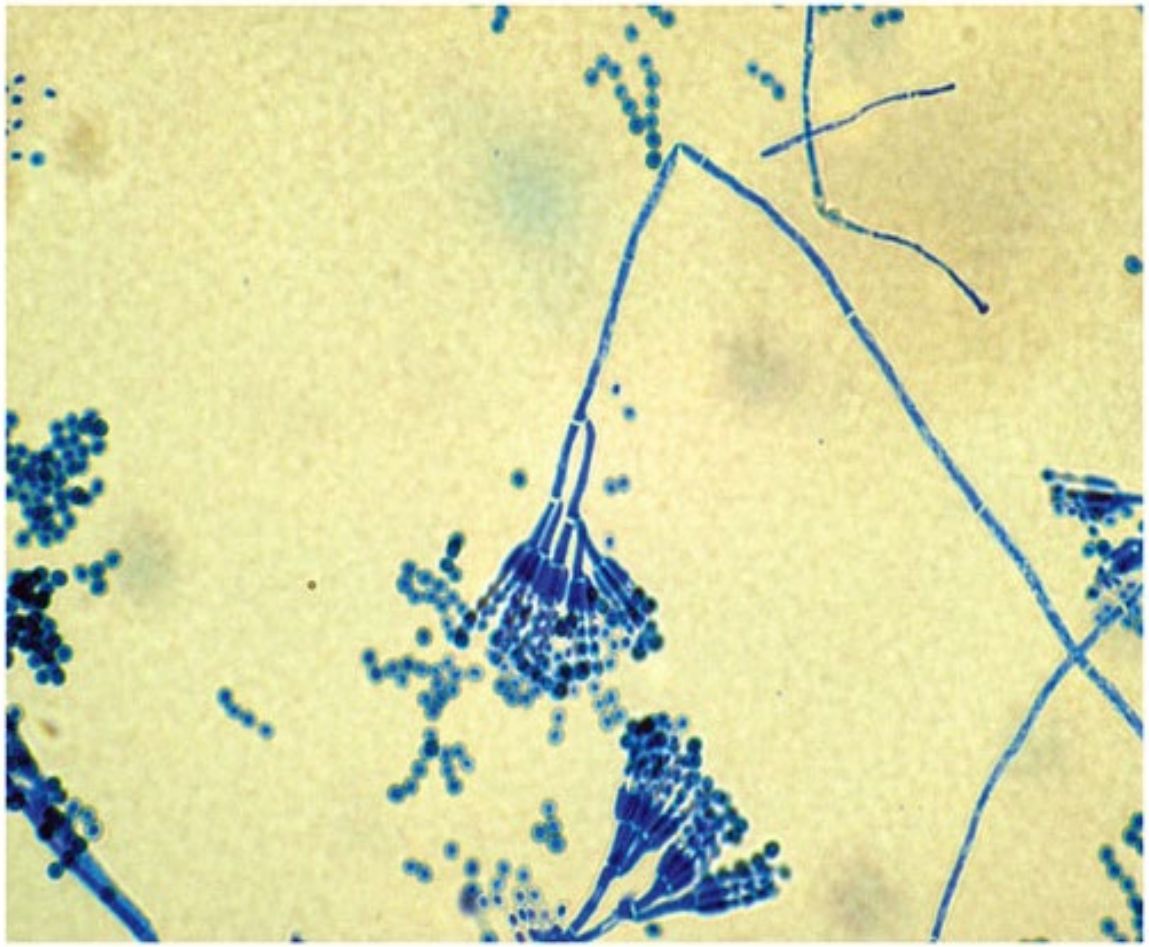


Image from: Mikotoksyny i grzyby pleśniowe. Zagrożenia dla człowieka i zwierząt (Wydawnictwo Uniwersytetu Kazimierza Wielkiego, 2006).

In the early 1970s, the Japanese biochemist Akira Endo and his colleagues suspected that some fungi might produce compounds that inhibit one or more of the enzymes in the biochemical pathway that produces cholesterol. They screened **6,000 microbial strains** in search of such a compound.

Penicillium citrinum, a relative of the fungus that makes blue cheese blue, yielded **mevastatin**, which was the first cholesterol-lowering compound to be identified.

Endo's discovery of mevastatin prompted other scientists and large pharmaceutical companies to seek similar compounds.

Cholesterol synthesis: requirements



The synthesis of cholesterol is costly:

18 acetyl-CoA

18 ATP

16 NADPH

Cholesterol synthesis: regulation

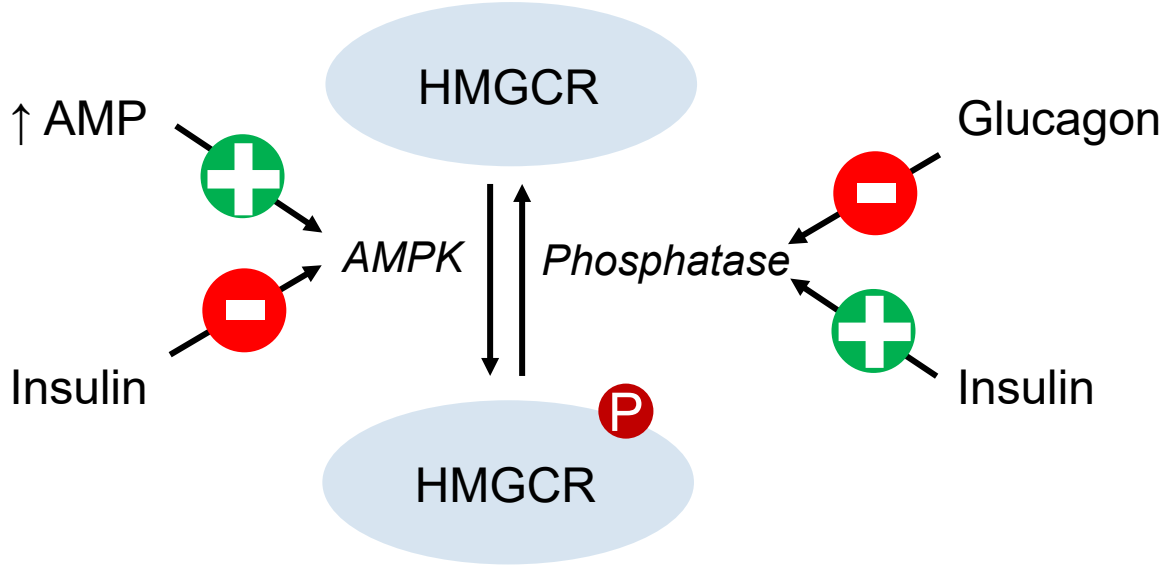
Regulation of *HMG-CoA reductase*

i. Transcriptional regulation
ii. Sterol-dependent degradation

Long-term

iii. Reversible phosphorylation

Short-term

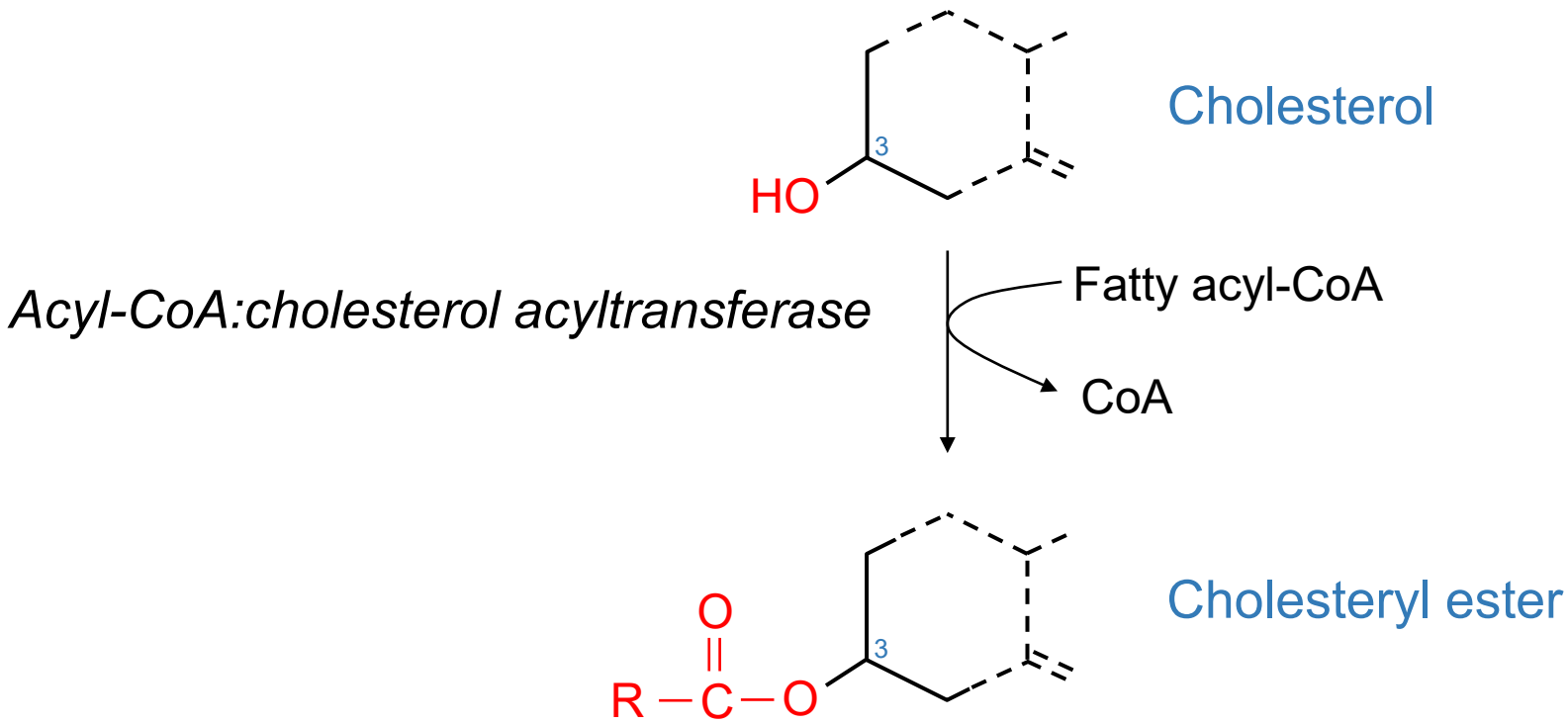


Fate of cholesterol molecules

- The liver is a major site of cholesterol synthesis
- Hepatic cholesterol can be:
  - converted into bile acids
  - secreted into bile as free cholesterol
  - esterified for storage or lipoprotein transport
- Other tissues use cholesterol in membranes or convert it into steroid hormones or vitamin D.

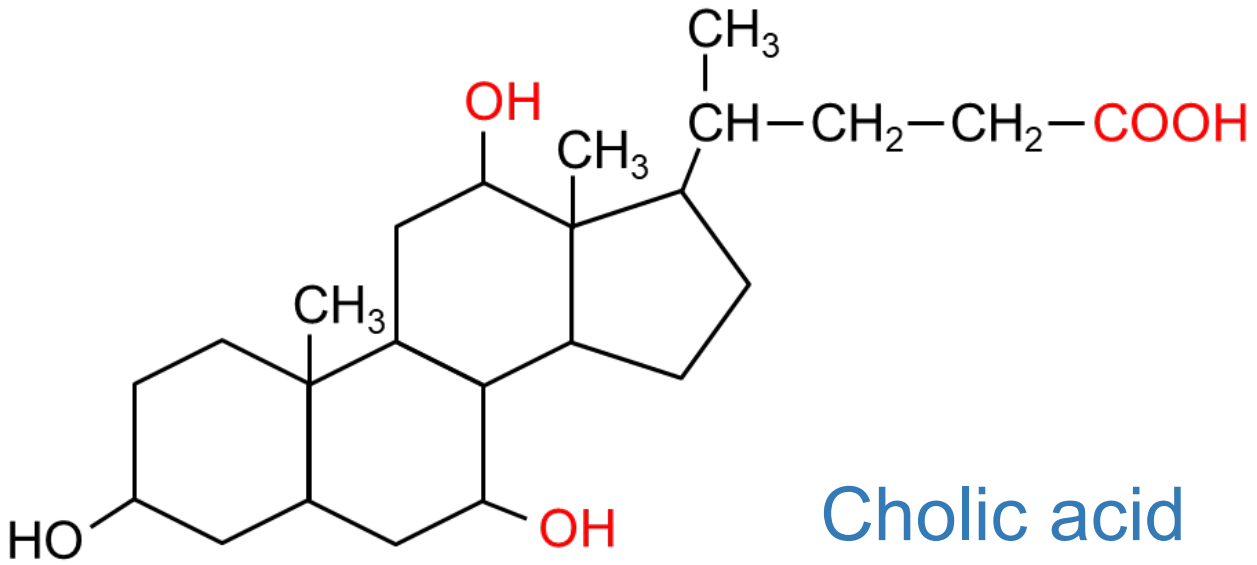
Cholesteryl esters

- ACAT catalyzes cholesterol esterification
- Cholesteryl esters are used for storage and transport



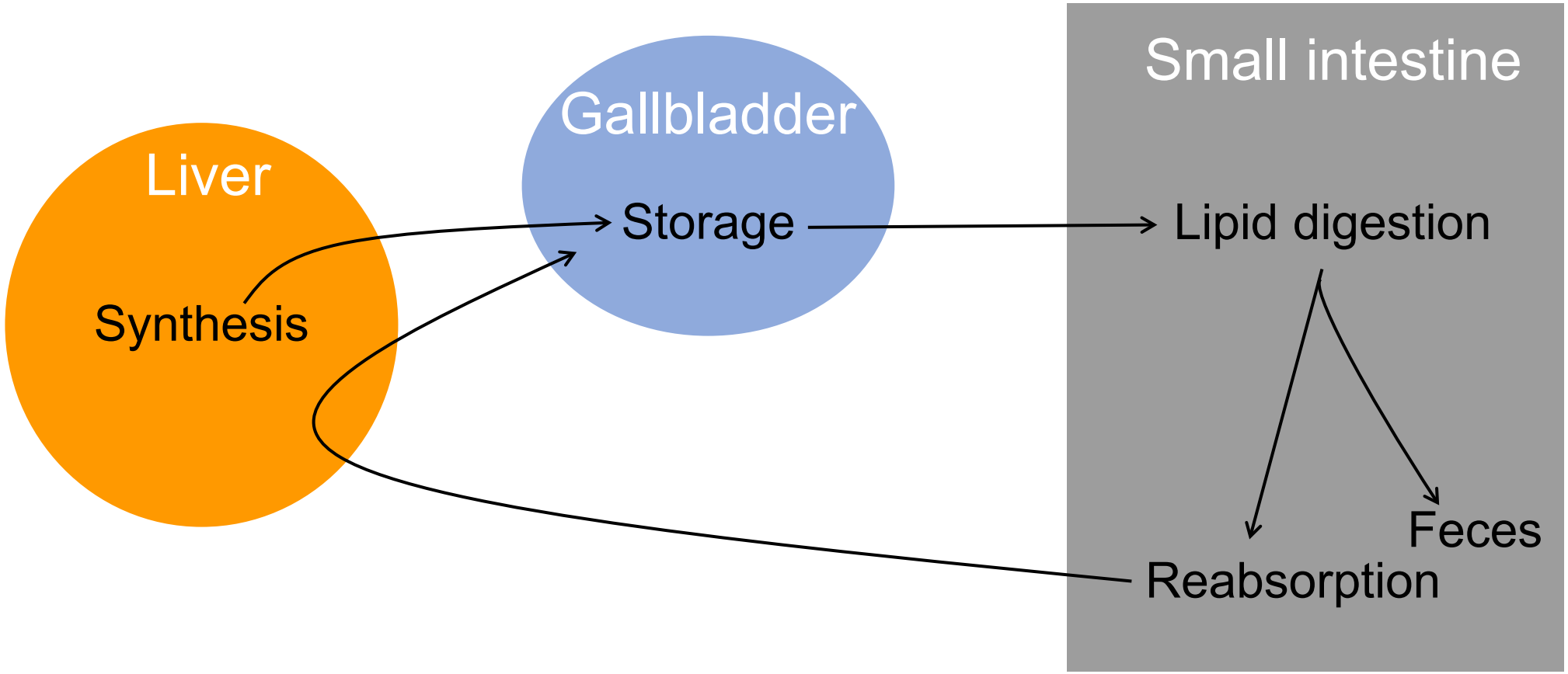
Bile acids

- Bile acids are synthesized in the liver, stored in the gallbladder, and secreted into the small intestine after a fatty meal
- Bile acids facilitate the emulsification and absorption of dietary lipids

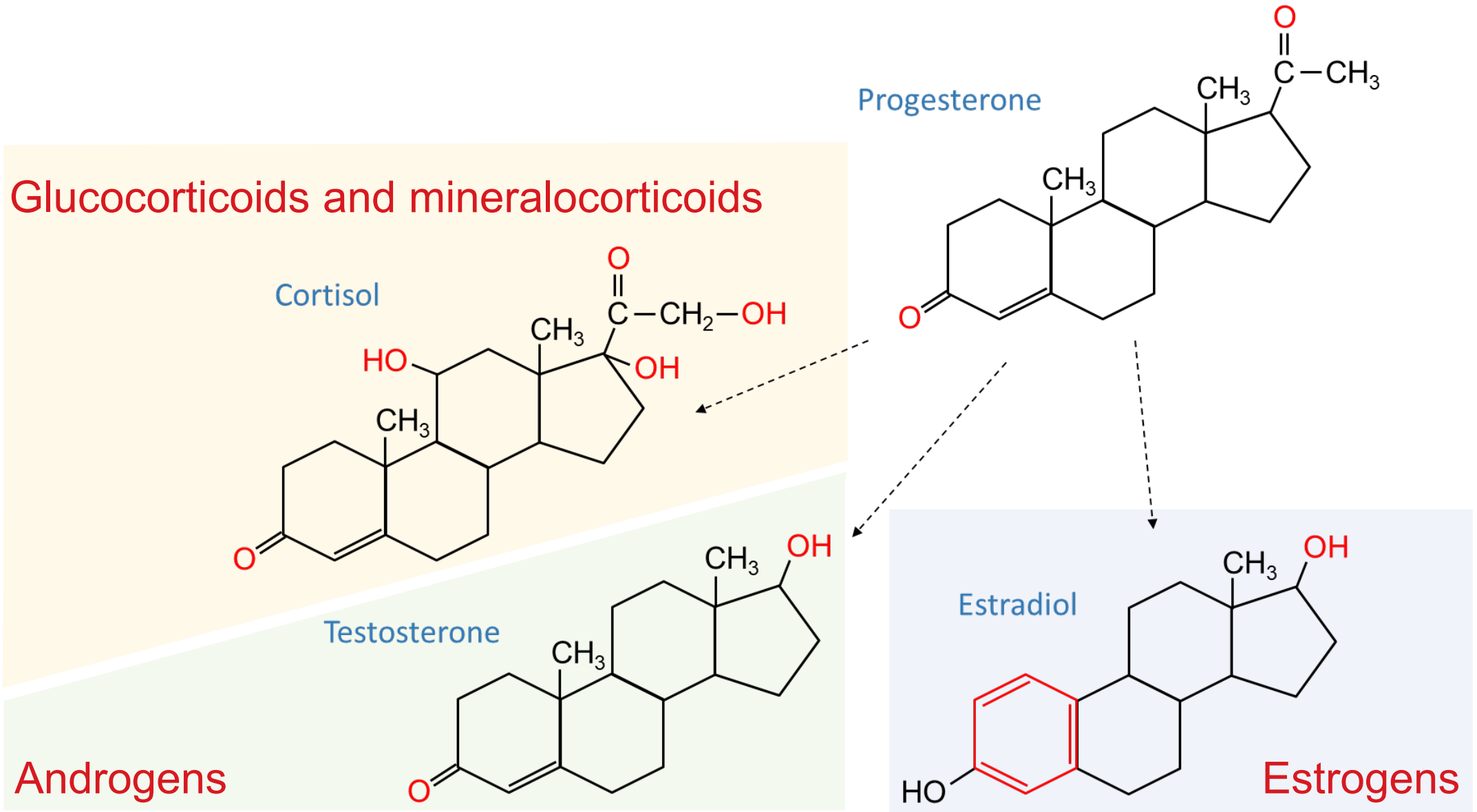


Cholic acid

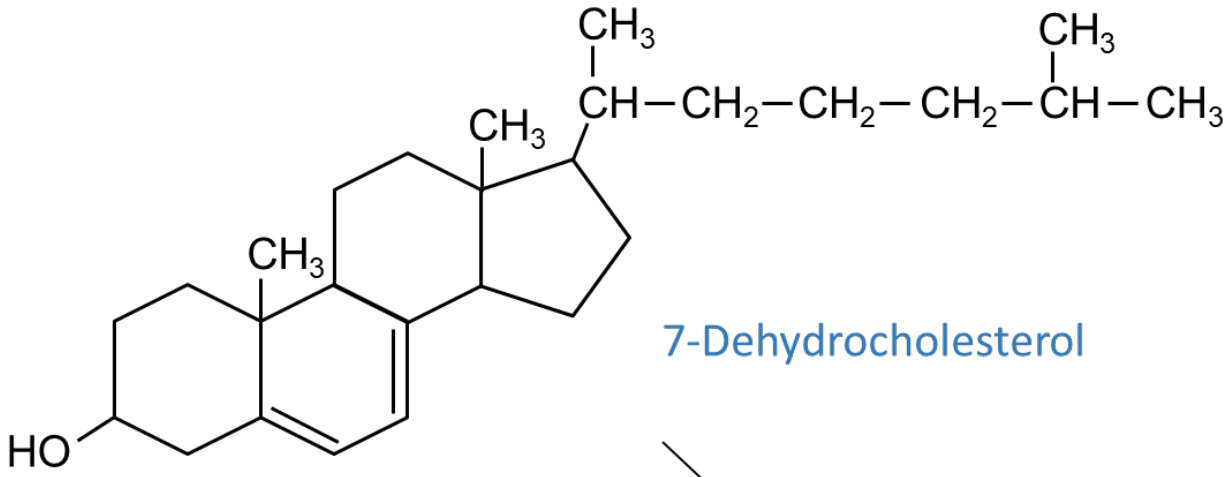
Enterohepatic circulation



Steroid hormones

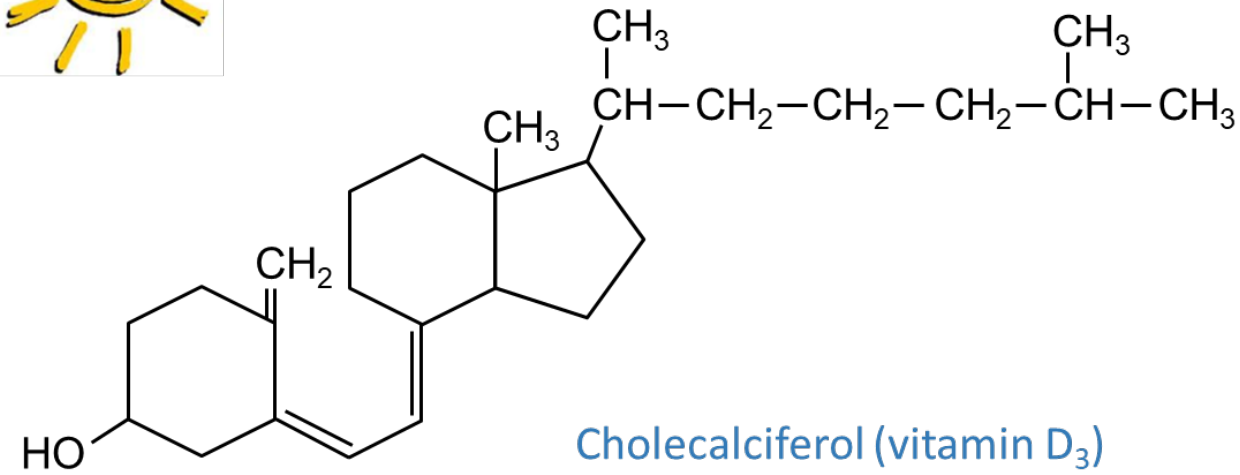


Vitamin D_3 synthesis in the skin



7-Dehydrocholesterol

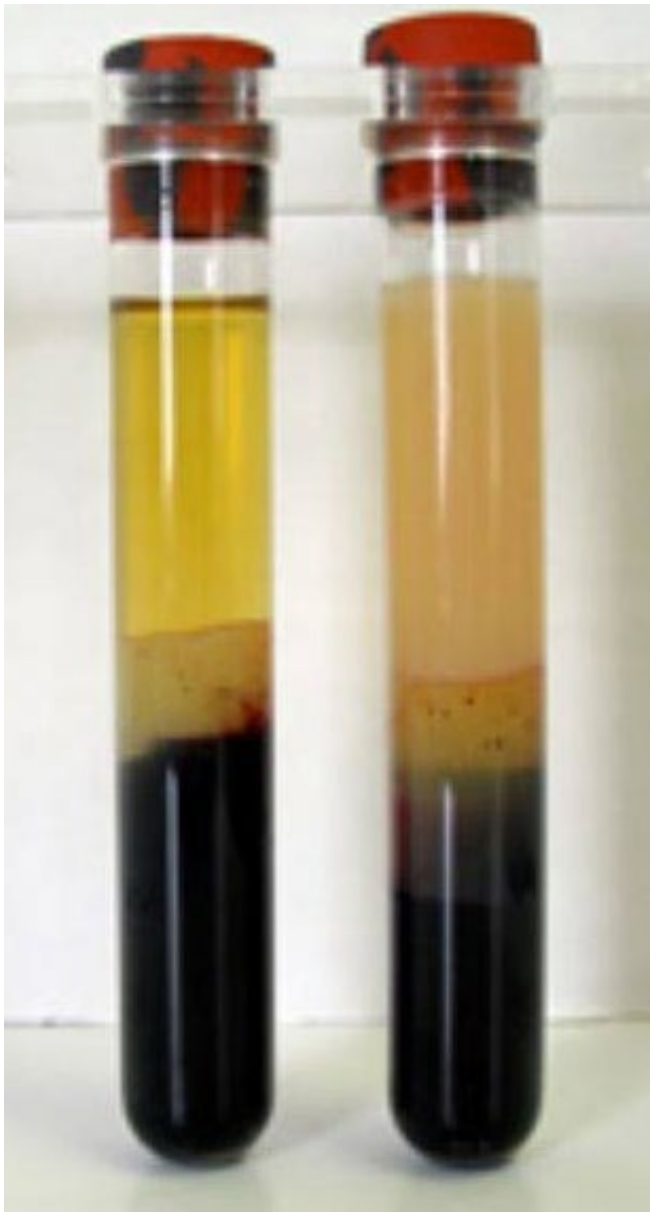




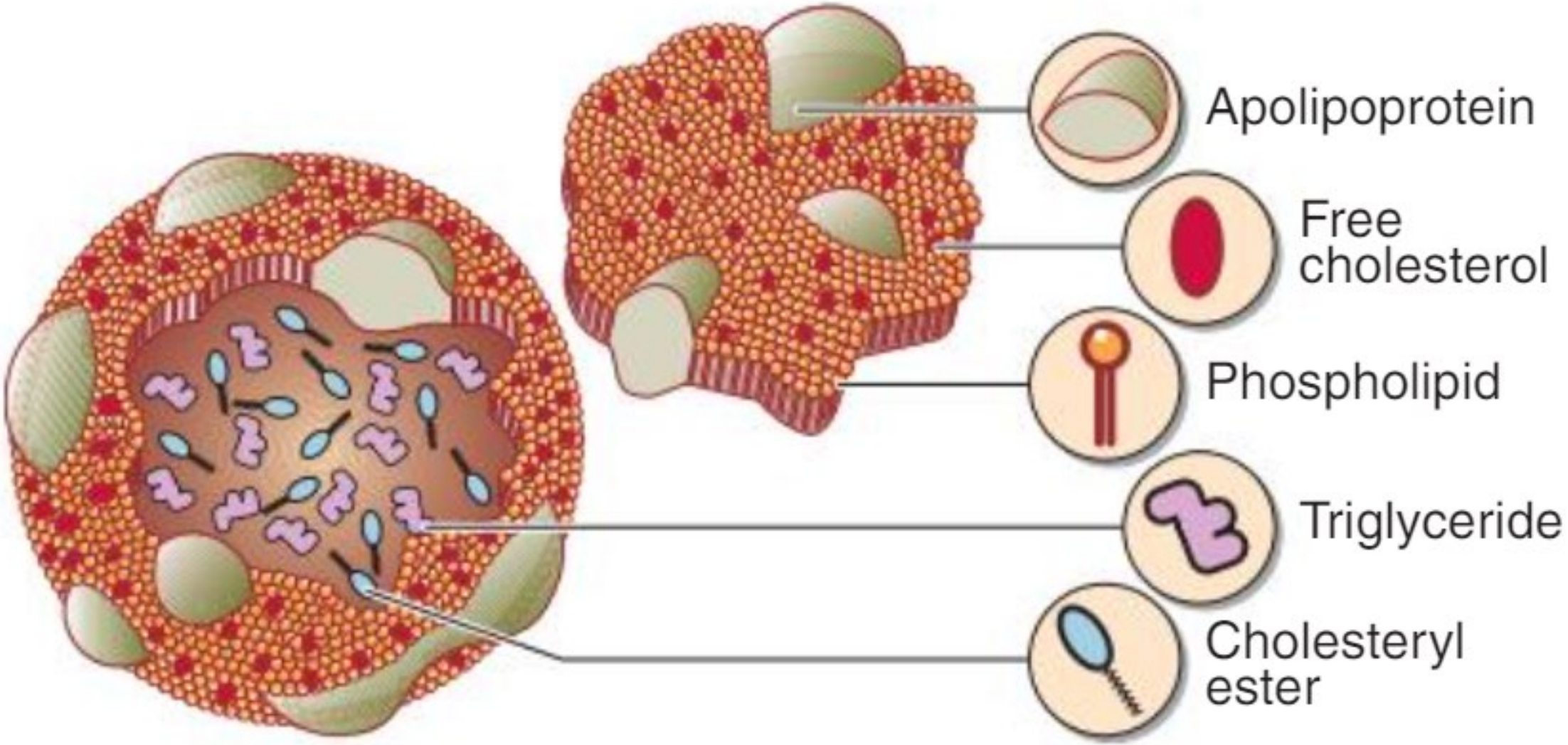
Cholecalciferol (vitamin D_3)

Transport of lipids





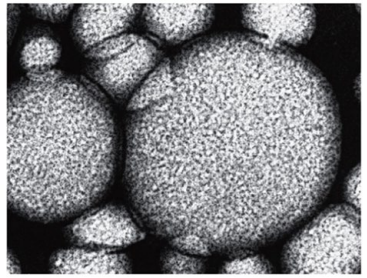
Lipoproteins



Murphy's Clinical Biochemistry 6th ed.

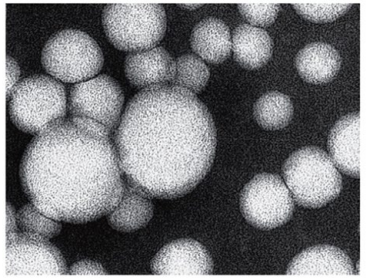
Four main classes of lipoproteins

- Chylomicrons



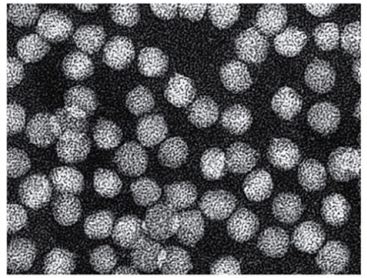
Chylomicrons (×60,000)

- Very low-density lipoproteins (VLDL)



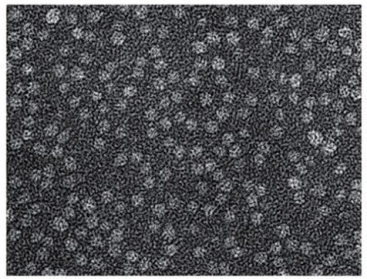
VLDL (×180,000)

- Low-density lipoproteins (LDL)



LDL (×180,000)

- High-density lipoproteins (HDL)



HDL (×180,000)

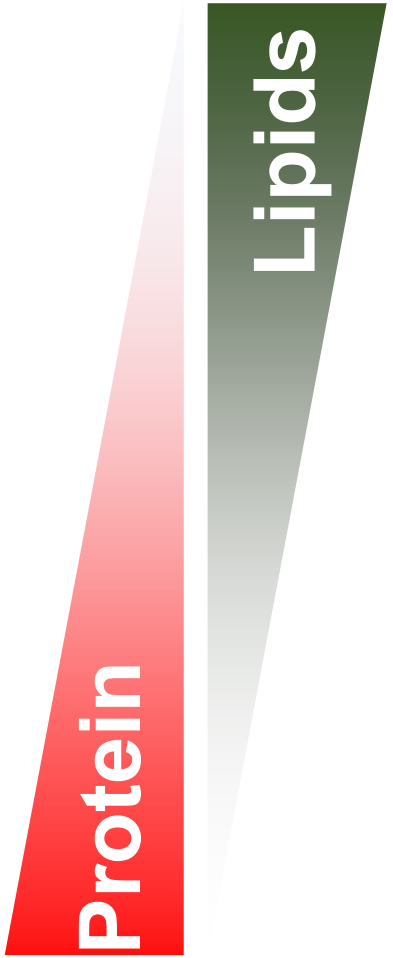
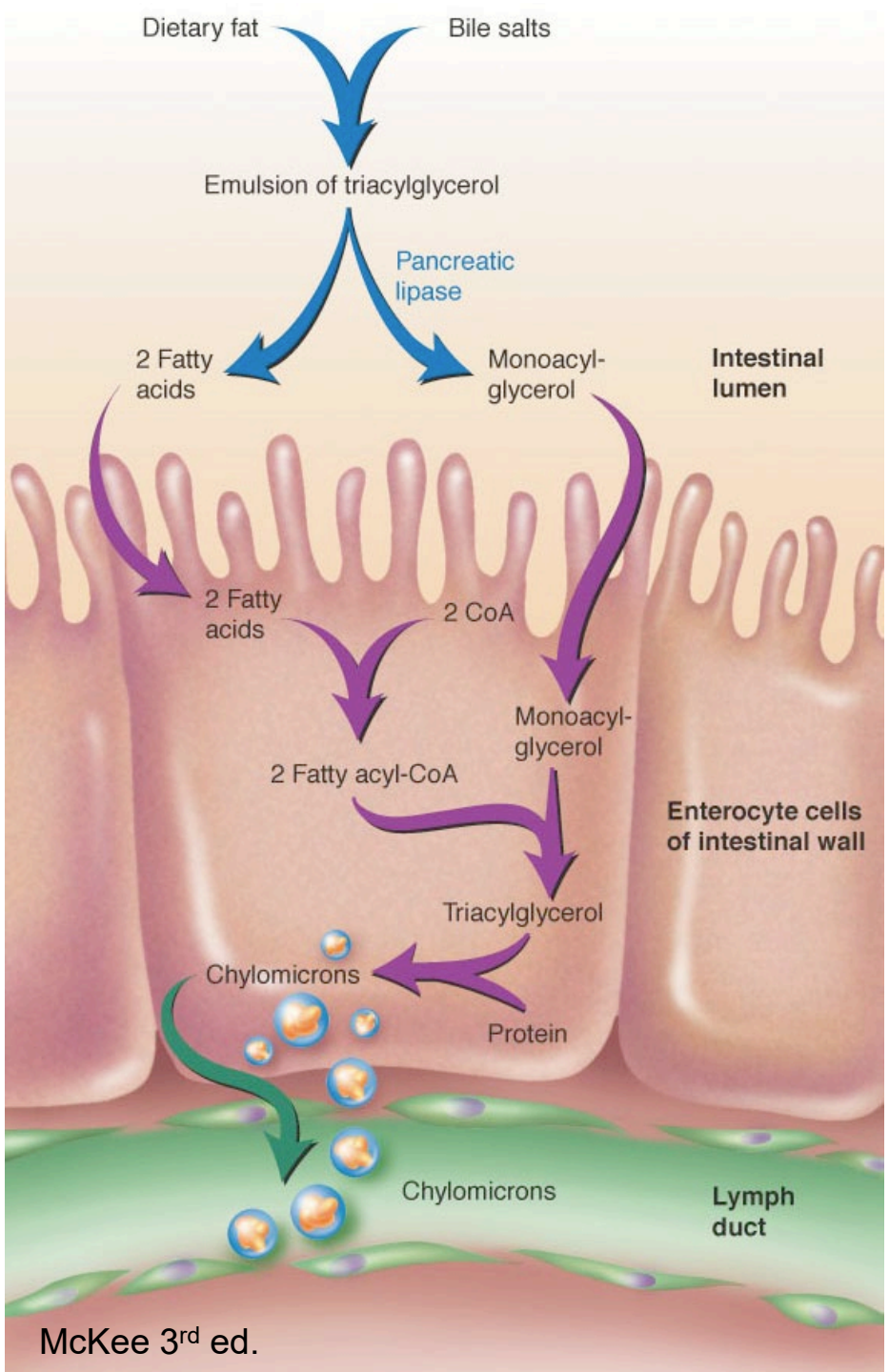


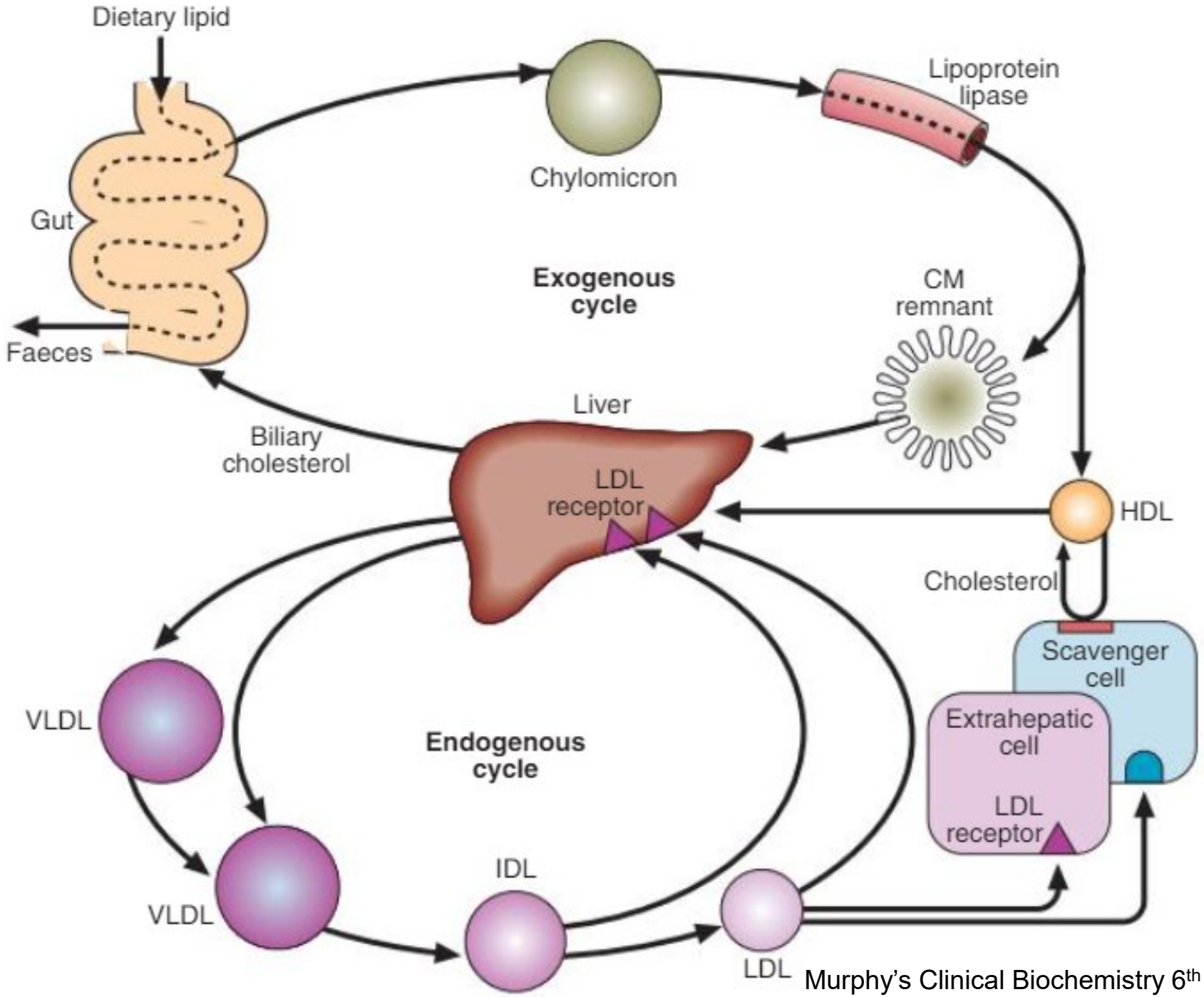
Figure 21-39b
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Lipid digestion and absorption



McKee 3rd ed.

Functions of lipoproteins



Murphy's Clinical Biochemistry 6th ed.

Functions of lipoproteins

Table 67.2 **The four main lipoproteins and their functions**

Lipoprotein	Main apolipoproteins	Function
Chylomicrons	B_{48}, A-I, C-II, E	Largest lipoprotein. Synthesised by gut after a meal. Not present in normal fasting plasma. Main carrier of dietary triglyceride.
Very-low-density lipoprotein (VLDL)	B_{100}, C-II, E	Synthesised in the liver. Main carrier of endogenously-produced triglyceride
Low-density lipoprotein (LDL)	B_{100}	Generated from VLDL in the circulation. Main carrier of cholesterol
High-density lipoprotein (HDL)	A-I, A-II	Smallest lipoprotein. Protective function. Takes cholesterol from extrahepatic tissues to the liver for excretion (reverse cholesterol transport).

Murphy's Clinical Biochemistry 6th ed.

Apolipoproteins

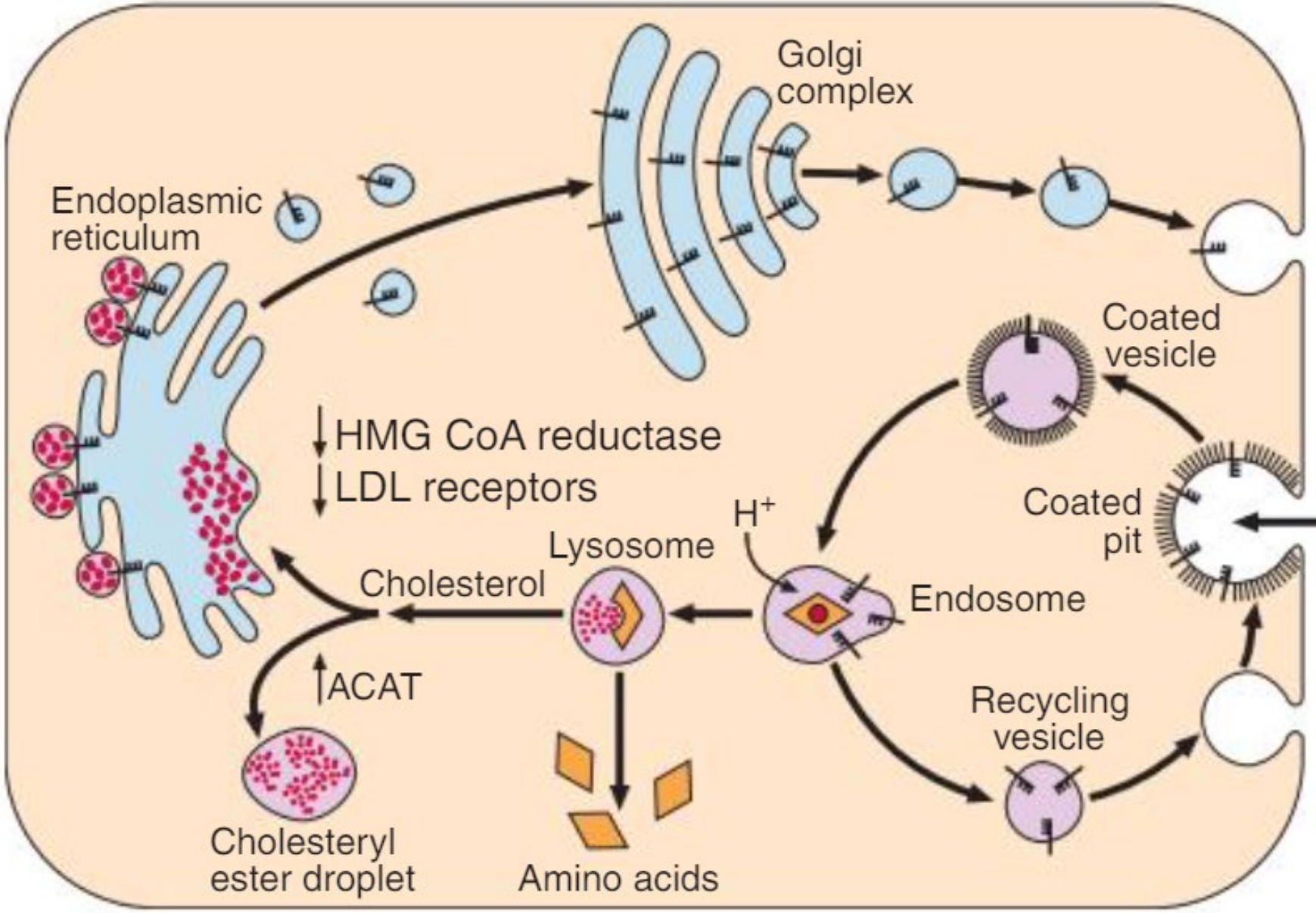
Table 67.1 **Properties of some human apolipoproteins**

Apolipoprotein	Molecular weight	Site of synthesis	Function
A-I	28 000	Intestine, liver	Activates LCAT
A-II	17 000	Intestine, liver	–
B_{100}	549 000	Liver	Triglyceride and cholesterol transport. Binds to LDL receptor
B_{48}	264 000	Intestine	Triglyceride transport
C-I	6600	Liver	Activates LCAT
C-II	8850	Liver	Activates LPL
C-III	8800	Liver	? Inhibits LPL
E	34 000	Liver, intestine, macrophage	Binds to LDL receptor and probably also to another specific liver receptor

LCAT = Lecithin–cholesterol acyltransferase.
LPL = Lipoprotein lipase.

Murphy's Clinical Biochemistry 6th ed.

LDL cholesterol uptake



Clathrin-mediated endocytosis

Murphy's Clinical Biochemistry 6th ed.

Cell, Vol. 10, 351–364, March 1977, Copyright © 1977 by MIT

Role of the Coated Endocytic Vesicle in the Uptake of Receptor-Bound Low Density Lipoprotein in Human Fibroblasts

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